

# THE BONDI DIPOLE IN FULL NUMERICAL RELATIVITY: A SELF-ACCELERATING POSITIVE–NEGATIVE MASS BINARY WITH GRTECLYN

NIKITA M. SHIROKOV\*

*Independent Researcher, Moscow, Russia*

(Dated: August 15, 2026)

Bondi observed in 1957 that a pair of bodies whose active gravitational masses have opposite signs should *self-accelerate*: the negative body is attracted by the positive one, the positive body is repelled by the negative one, and the pair runs off together. Seven decades of point-particle and exact-solution analyses never posed this configuration as an initial-value problem with dynamical matter. Here we do, presenting the first full 3+1 numerical-relativity evolutions of a “Bondi dipole”—built on what is, to our knowledge, the first stable star with negative total (ADM) mass evolved in numerical relativity. The model is minimal: two complex scalar fields obey the *same* sextic Klein–Gordon dynamics, and only the phantom sector enters the Einstein equations with a flipped sign, so its inertial and passive masses stay positive while its active mass is negative. Both stars are solved as constraint-satisfying initial data with residuals below 0.1%. Released at rest, the mixed pair co-drifts at 50 to 150 times the noise floor set by its single-star and same-sector controls; mirroring the configuration reverses the trajectory to a few percent; the canonical star’s displacement scales with the phantom’s active mass to 0.6%; and at initial separations of 12 and 16 the early-time drift is Newtonian to about 10%, the excess at separation 8 being a measured extended-body overlap effect. In the signed bookkeeping the phantom sector carries negative momentum, so the runaway is consistent with momentum conservation, as in Forward’s Newtonian analysis.

## I. INTRODUCTION

Bondi [1] gave negative mass its modern treatment, distinguishing inertial, passive, and active gravitational mass and noting the striking consequence of mixing signs: a negative-mass body is *attracted* by a positive one, the positive body is *repelled* by the negative one, and the pair runs off together—momentum conserved, because the negative body’s contribution is negative. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated opposite-mass pairs, later generalized in Ref. [3]; Bonnor [4] asked why mass is positive and found no answer in the theory. Forward [5] analyzed the Newtonian two-body problem exhaustively—for equal and unequal mass magnitudes alike—showing “in tedious detail” that the self-accelerating pair violates neither momentum nor energy conservation; objections to negative mass must be sought elsewhere [6, 7].

All of this work treats point particles, prescribed metrics, or frozen mass parameters. It cannot say whether such a configuration can be posed as a *consistent initial-value problem*—constraint-satisfying initial data for dynamical matter—or whether the runaway survives full nonlinear evolution with backreaction, radiation, and finite body size. That is the gap this paper closes; we are not aware of any previous 3+1 evolution of a gravitationally bound positive–negative active-mass pair with dynamical, constraint-solved matter.

Our contributions:

- (i) A minimal field-theoretic realization of Bondi’s sign structure: a *bicomplex* scalar model in which both

sectors obey identical Klein–Gordon dynamics and only the Einstein source is signed (Sec. II).

- (ii) Dressed soliton stars for both signs, solved at fixed frequency as constraint-clean initial data—including, as far as we know, the first negative-ADM-mass (“phantom”) star evolved this way (Sec. III).
- (iii) A falsifiable run matrix—mixed pair, same-sector pairs, single stars, a mirrored pair, two wider separations—run under identical numerics with birth-time verification (Secs. IV, V).
- (iv) The result: the mixed pair, and only the mixed pair, self-accelerates—with unequal mass magnitudes (the reference pair,  $|M_-|/M_+ = 1.20$ ) and with the magnitudes matched, echoing Forward’s unequal-mass cases; the effect scales with the source mass and converges to the Newtonian point-mass prediction as the bodies separate (Sec. V).
- (v) An honest accounting of failure modes and diagnostic biases that mimic or mask the physics under test (Secs. IIID, VI).

Throughout,  $G = c = 1$  and the scalar mass sets the unit,  $m = 1$ . Drifts along the pair axis are quoted as  $\Delta x = x(t) - x(0)$ , and  $d$  denotes the instantaneous barycentre separation. The paper follows the build order: Sec. II defines the matter model, Sec. III constructs the stars and the initial data—including the failed attempts that shaped the recipe—Sec. IV lays out the runs, Sec. V the measurements, and Secs. VI–VII their limits and meaning.

---

\* shirokov.nm@phystech.edu

## II. THE BICOMPLEX SCALAR MODEL

We first need matter that can realize Bondi’s sign structure while remaining an honest field theory. The minimal choice: two independent complex scalar fields,  $\Phi_+$  (*canonical*) and  $\Phi_-$  (*phantom*), living on one spacetime and sharing one potential,

$$V(|\Phi|^2) = \frac{1}{2}m^2|\Phi|^2 - \frac{1}{4}\lambda|\Phi|^4 + \frac{1}{6}\mu|\Phi|^6, \quad (1)$$

with  $\lambda, \mu > 0$ . The sextic term is required—the attractive quartic alone is critical in three dimensions and either collapses or disperses—and gives the flat-space theory Q-ball-type solitons [9, 10]: localized, self-bound lumps of field. (“Lump” is our generic word for such an object; it becomes a “star” once gravity dresses it.)

The entire sign structure lives in one place: the Einstein equations are sourced by the *signed* sum of two canonical stress tensors,

$$G_{\mu\nu} = 8\pi (T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]), \quad (2)$$

while the field equations of *both* sectors keep their canonical form,

$$\square \Phi_{\pm} = \frac{\partial V}{\partial |\Phi|^2} \Phi_{\pm}. \quad (3)$$

In the 3+1 language the matter sources seen by normal observers—the energy density  $\rho$  and momentum density  $S_i$  that feed the ADM constraint and evolution equations—are accumulated per sector with a sign  $\epsilon_{\pm} = \pm 1$ :

$$\rho = \sum_{s=\pm} \epsilon_s \rho[\Phi_s], \quad S_i = \sum_{s=\pm} \epsilon_s S_i[\Phi_s], \quad (4)$$

and likewise for  $S_{ij}$ , each bracket being the standard scalar-field expression.

The consequences follow Bondi’s premise exactly. Because Eq. (3) is canonical, the phantom field propagates like ordinary matter—*inertial and passive* masses positive, falling toward wells. Because of the sign in Eq. (2), its *active* mass is negative—and with it its Arnowitt–Deser–Misner (ADM) mass, the total mass an observer at spatial infinity would read off the far field: it repels everything. A mixed pair therefore chases, the phantom falling after the canonical star it repels, both accelerating the same way. Newton’s third law fails for the two bodies in the plain sense that the two forces point the *same* way rather than opposite ways; the bookkeeping is nonetheless sound: the conserved quantities belong to the total signed stress-energy, in which the phantom’s momentum enters with a minus sign, so the co-moving pair carries near-zero total momentum—Forward’s Newtonian cancellation [5], general-relativistically.

Two honest remarks. The phantom sector violates the null energy condition by construction; this is a model of *if such matter existed, what would gravity do*, in the spirit of phantom cosmology [8], not a claim that it exists. And

our sign structure differs from Forward’s all-four-masses-negative matter [5]: here only the source sign flips. The two-body phenomenology is the same, because passive and inertial mass cancel—each body’s acceleration is set by the *other* body’s active mass, the property under test.

The model is implemented in `GRTEclyn` [15, 16, 18], evolving CCZ4 [14] plus the two complex fields, with Eq. (4) accumulated star by star in the initial-data solver `GRtresna` [17] so that one constraint solve can carry mixed-sign matter.

## III. INITIAL DATA: GRAVITATIONALLY DRESSED SOLITONS

The task of this section is to turn the model into two stars on one initial slice: fix the couplings, solve each star as an equilibrium, stamp (“paint”) the solutions onto the evolution grid, and solve the constraints for the pair. The failed campaigns that forced each of these choices close the section.

### A. Couplings

The potential (1) has three parameters: the mass  $m = 1$  (our unit) and the self-couplings  $\lambda, \mu$ . Two combinations control the physics. The dimensionless ratio  $\lambda^2/\mu$  sets the *shape* of the potential and thereby the flat-space frequency floor  $\omega_{\min} = \sqrt{1 - 3\lambda^2/16\mu}$  below which no soliton exists; all runs keep  $\lambda^2/\mu = 4.8$ , so  $\omega_{\min} = 0.316$  and the working frequency  $\omega = 0.55$  always means 45% binding. At fixed shape, the overall coupling strength sets the soliton amplitude through the thin-wall value  $\sqrt{3\lambda/4\mu}$  (the interior amplitude of a large flat-space soliton), and with it how strongly the lump gravitates. The campaign settled on  $\lambda = 10240$ ,  $\mu \simeq 2.18 \times 10^7$  (amplitude  $\simeq 0.019$ )—the *ultra*weak rung of a ladder descended from stronger couplings whose stars would not hold together (Sec. III D), and the first rung at which a dressed  $\omega = 0.55$  star exists at all.

### B. The dressed-star problem

Dressing the soliton in its own gravity is the boson-star problem [11, 12] with a sextic self-interaction (see Ref. [13] for a review), posed for both source signs. Inserting the harmonic ansatz  $\Phi = \varphi_0(r) e^{-i\omega t}$  and a conformally flat, maximally sliced metric  $ds^2 = -\alpha^2 dt^2 + \psi^4 \delta_{ij} dx^i dx^j$  into Eqs. (2)–(3) reduces the problem to three coupled ordinary differential equations in  $r$  (the

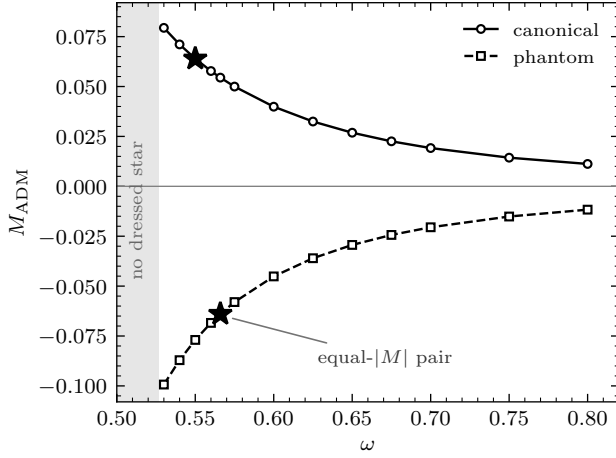


FIG. 1. The dressed-star family  $M_{\text{ADM}}(\omega)$  at the ultraweak rung. At equal frequency the phantom star is the heavier in magnitude (20% at  $\omega = 0.55$ , 4% at  $\omega = 0.80$ ): repulsive self-gravity puffs it out. The branch ends near  $\omega \approx 0.52$ – $0.53$  (shaded). Stars (\*): the equal- $|M|$  pair evolved for the mass-scaling test, matched to one part in  $10^5$ .

standard boson-star reduction [13]),

$$\begin{aligned} \psi'' + \frac{2}{r}\psi' &= -2\pi\psi^5\rho, \\ \alpha'' + \left(\frac{2}{r} + \frac{2\psi'}{\psi}\right)\alpha' &= 4\pi\psi^4\alpha(\rho + S), \\ \varphi_0'' + \left(\frac{2}{r} + \frac{\alpha'}{\alpha} + \frac{2\psi'}{\psi}\right)\varphi_0' &= \psi^4\left(\frac{dV}{d\varphi_0} - \frac{\omega^2}{\alpha^2}\varphi_0\right), \end{aligned} \quad (5)$$

with  $\rho = \frac{1}{2}\omega^2\varphi_0^2/\alpha^2 + \frac{1}{2}\psi^{-4}\varphi_0'^2 + V$  and  $\rho + S = 2\omega^2\varphi_0^2/\alpha^2 - 2V$ . **The phantom star is obtained by flipping the sign of  $\rho$  and  $\rho + S$  in the metric equations only;** the Klein–Gordon line keeps its canonical form, mirroring Eqs. (2)–(3). The result is a self-interaction-bound soliton with repulsive gravity:  $\psi < 1$ ,  $\alpha > 1$  in the core (reversed for the canonical star) and negative ADM mass.

We integrate Eqs. (5) by shooting; the solver and the  $M(\omega)$  family scan behind Fig. 1 ship in the results pack. Solving at *fixed frequency* matters: heavy-branch sextic stars sit a fraction of a percent below the effective-potential top, so at fixed central amplitude the eigenvalue in  $\omega$  is an exponentially thin needle, and a shooter on  $\omega$  lands on the light branch (a request for the  $\omega = 0.55$  star returned a  $5\times$  lighter  $\omega = 0.75$  star). Instead we bisect the central amplitude  $\varphi_c$  at fixed integration-frame frequency and outer-iterate until the rescaled frequency  $\omega/\alpha_\infty$  matches the request, to  $\sim 10^{-5}$ .

Figure 1 shows the family. The evolved stars: canonical  $\omega = 0.550$  ( $\varphi_c = 0.019695$ ,  $M_{\text{ADM}} = +0.063951$ ,  $\alpha(0) = 0.9773$ ); phantom  $\omega = 0.550$  ( $\varphi_c = 0.019589$ ,  $M_{\text{ADM}} = -0.076962$ ,  $\alpha(0) = 1.0249$ ;  $|M_-|/M_+ = 1.203$ ); and, for the equal- $|M|$  run of Sec. IV, phantom  $\omega = 0.56598$  ( $M_{\text{ADM}} = -0.063950$ ).

### C. Painting and constraint solve

Two seeding subtleties mattered enormously (Sec. III D). The seed amplitude must be the star’s own solved  $\varphi_c$ —any rescaling de-tunes the eigenstate. And a boson star is stationary, not static: its phase rotates, so the field carries a  $U(1)$  momentum, and the momentum conjugate is defined per unit *proper* time of the normal observers,  $\Pi = -\alpha^{-1}\partial_t\Phi$  for zero shift. The harmonic phase therefore seeds

$$\Pi_{\text{im}} = -\frac{\omega}{\alpha(r)}\varphi_0(r); \quad (6)$$

a flat-space seed ( $\alpha \equiv 1$ ) would plant the star slightly off its eigenstate everywhere the lapse differs from one, which is why the star tables carry  $\alpha(r)$ . Each sector gets its own profile table, with the frequency threaded through per lump—painting the  $\omega = 0.56598$  profile with a global  $\omega = 0.550$  phase velocity would seed it off-eigenstate.

**GRtresa** solves the constraints (CTTKHybrid, per-lump signed sources) to 0.1%; at the earlier 1% gate the residual radiated away as a  $\chi$  ring that sloshed the stars’ envelopes. The evolution keeps the solved metric and repaints the matter analytically from the same tables.

### D. How the matter was made to hold together

None of the choices above were right on the first try. Before any runaway could be claimed, a single lump had to hold its own size; four campaigns failed that gate. Heavy lumps collapsed ( $\min \chi \rightarrow 0.009$  by  $t \approx 25$ ), light lumps evaporated ( $\text{rms } 5 \rightarrow 30$  by  $t \approx 50$ ), and the geometric-mean rung did both—indicting the *shape* of the seed, not its weight. Four root causes, in order:

1. **The amplitude clamp.** The seed pipeline—the script that stamps a star’s profile onto the grid—clamped the amplitude to the flat-space thin-wall value  $\sqrt{3\lambda/4\mu}$ , 4.5% below the solved eigenstate: an exact Q-ball shape at the wrong height is not a Q-ball, and it breathed, shed, and evaporated. *Fix:* seed at the solved  $\varphi_c$ .
2. **No dressed equilibrium existed.** The corrected flat-space seed still dispersed: a flat-space eigenstate is an equilibrium only in flat space, and the dressed family showed the target star did not exist at the original couplings. *Fix:* move to the ultraweak rung and seed the dressed profile with its own lapse, Eq. (6).
3. **Two painters disagreed silently.** The constraint solve used the tabulated star while the evolution’s repaint fell through to a Gaussian: the solved metric backed one object, the evolution started another. *Fix:* both paths resolve the same star, enforced to  $10^{-10}$  by a regression test.

4. **The phantom path had never run.** An old veto downgraded exotic lumps to canonical matter whenever self-gravity was on—correct for a gravity-bound star, irrelevant for one bound by self-interaction. *Fix:* lift the veto. The first phantom star evolved hit its predicted fingerprint, with the predicted mirror-image metric signature ( $\chi$  just above unity).

Two residual effects were attributed rather than fixed: the  $\pm 8\%$  radial breathing is resolution-intrinsic (resolving with  $30\times$  tighter momentum residuals reproduced it crest for crest), and the rms-radius stream tracks the shed radiation bath (Sec. VI), not star health.

The exportable lesson: *verify the seed before believing the evolution.* Every run was checked at  $t = 0$  against an independently integrated fingerprint; two of the four root causes were caught by that check, not by watching movies.

#### IV. EVOLUTION SETUP AND THE RUN MATRIX

With both stars in hand, the experiment is to release them at rest and watch the barycentres. Evolutions use `GRTEclyn` (one GPU per run, single rank): CCZ4 [14] with damping  $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$ ,  $1 + \log$  slicing and a Gamma-driver shift with  $\eta = 1$ , fourth-order spatial stencils, RK4 time stepping, Kreiss–Oliger dissipation  $\sigma = 2$ , and  $\Delta t = 0.02 \Delta x$ . The domain is  $L = 64$  at base resolution  $N = 128$  ( $\Delta x = 0.5$ ), with Sommerfeld (radiative) outer boundaries and wave-extraction shells at  $R = 8$  and 16. One mesh-refinement level is allowed (`max_level` = 1,  $\chi$ -gradient tagging with threshold 0.02, regrid every 16 steps), but it first triggers at  $t \approx 48$ , after the measurement window, so every quantitative result was produced on the uniform grid (Appendix A). Stars are seeded at  $x = \pm 4$  about the domain centre ( $d_0 = 8$ , about  $1.6\times$  their rms radius of  $\sim 5$ ), *at rest*—no boosts, no velocities, no driving of any kind—so any drift is gravitational.

Runs are named by content: PM is the mixed canonical–phantom pair, PP and MM the same-sector pairs, P and M the single stars, MP the mirrored pair (sectors swapped in place), and PM-eq the mixed pair rerun with the phantom’s mass magnitude matched to the canonical star’s,  $|M_-| = M_+$  to one part in  $10^5$  (Fig. 1).

The primary diagnostic is the per-sector barycentre stream: field-weighted integrals, centroids, and rms radii for  $\Phi_+$  and  $\Phi_-$ , sampled continuously. Constraint norms (Appendix A), confinement (peak amplitude, confined fraction,  $\min \chi$ ),  $\Psi_4$  modes, and energy-condition monitors are recorded alongside.

Table I is the experiment: each row is a falsifiable prediction of Sec. II’s sign structure—a same-sector pair or lone star that drifted like the mixed pair would kill the claim. PM-eq changes exactly one physical quantity (the phantom’s mass, by 17%); MP reflects the geometry; the

TABLE I. The falsifiable run matrix. The sign structure of Sec. II predicts co-drift in  $+x$  for the mixed pair, reversal under mirroring, a weaker runaway at wider separation, and *no* net drift for same-sector pairs and lone stars. Every run was solved to Ham  $\leq 0.09\%$ , Mom  $\leq 0.08\%$  (Appendix A) and verified at  $t = 0$  against independent star fingerprints (within 0.2%). Drifts in code units at  $t_{\text{end}}$  ( $t = 40$  for singles, 60 for pairs).

| run                  | drift $\Phi_+$ | drift $\Phi_-$ | verdict        |
|----------------------|----------------|----------------|----------------|
| PM (+, −)            | +3.35          | +9.56          | <b>runaway</b> |
| PM-eq (equal $ M $ ) | +2.94          | +9.59          | runaway        |
| MP (mirror)          | −3.23          | −9.07          | reversed       |
| PP (+, +)            | +0.16          | —              | null           |
| MM (−, −)            | —              | +0.04          | null           |
| P (single +)         | +0.06          | —              | null           |
| M (single −)         | —              | +0.07          | null           |
| PM, $d_0=12$         | +0.72          | +2.78          | weaker         |
| PM, $d_0=16$         | (biased)       | +0.74          | weaker         |

wider separations probe the force law.

#### V. RESULTS

##### A. The runaway

Figures 2 and 3 carry the headline. In PM—canonical star at  $x = +4$ , phantom at  $x = -4$ , both at rest—*both* barycentres move in  $+x$ : by  $t = 60$  the canonical star has moved +3.35 and the phantom +9.56, closing the gap from 8.00 to 1.79. The four controls, run under identical numerics, do not move (+0.16 and +0.04 for the same-sector pairs, +0.06/+0.07 for the singles), setting the noise floor; the mixed pair’s drift exceeds it by 50 to 150 times, with constraint norms at the level of the non-drifting controls throughout the window (Appendix A).

The barycentre *speeds* (Fig. 4) say the same thing kinematically: both bodies accelerate together from rest—a sustained pull, not an impulse. The canonical star reaches  $v_x \approx 0.06$  at the window edge  $t = 30$  and peaks near 0.12; the phantom passes 0.11 at  $t = 30$  and reaches  $v_x \approx 0.44$ , a genuinely relativistic infall, as the pair nears contact. (The late decline of the canonical curve is the halo bias of Sec. VI, not a deceleration.)

Why does the phantom cover almost three times the distance? Not because point-mass gravity says so: each body is accelerated by the *other’s* active mass, and  $|M_-| > M_+$ , so the point-mass model actually gives the *canonical* star the larger early drift (Table II). The asymmetry is an extended-body effect, quantified in Secs. V C and V D: the push on the canonical star stays Newtonian in the phantom’s mass to 0.6%, while the phantom’s fall runs up to  $3\times$  ahead of the point-mass rate because its envelope sits deep inside the canonical star’s near field from  $t = 0$ —a well that deepens further as the canonical star

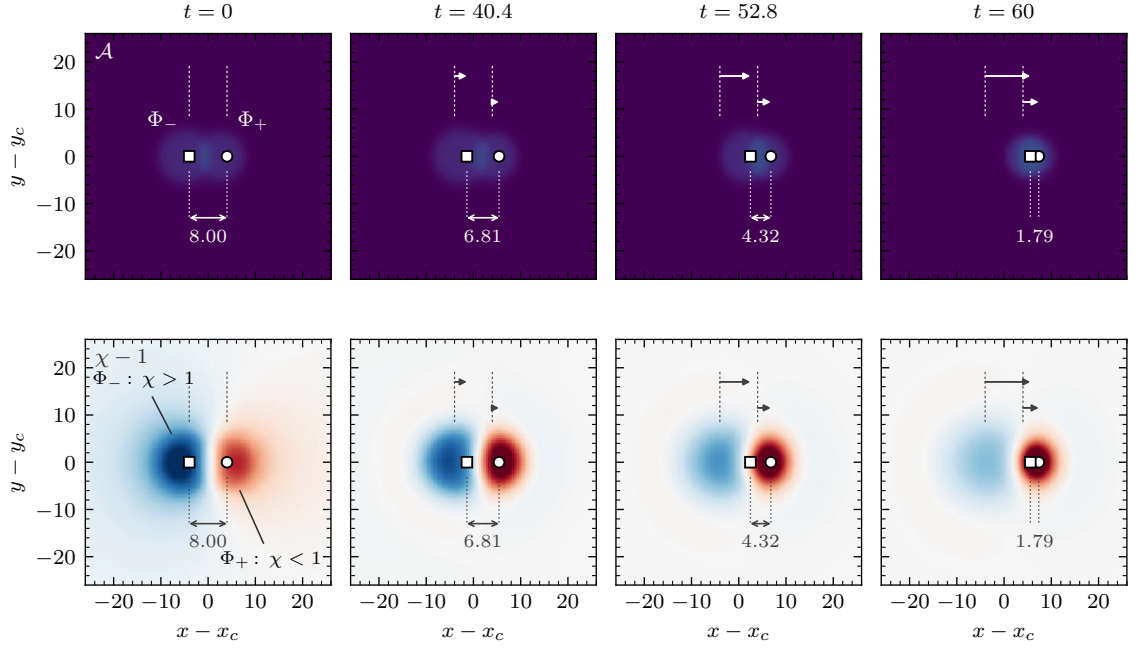


FIG. 2. The runaway, run PM, in the  $z = 0$  plane. Top: scalar-field activity  $\mathcal{A} = \sum_k (\varphi_k^2 + \Pi_k^2)^{1/2}$ , both sectors. Bottom:  $\chi - 1$ ; the phantom star is a hill ( $\chi > 1$ , blue), the canonical star a well ( $\chi < 1$ , red). Open markers: measured barycentres (circle canonical, square phantom); dashed verticals mark the release positions  $x = \pm 4$ , arrows the displacements; brackets give the separation,  $8.00 \rightarrow 1.79$ . Panels are individually color-scaled; the canonical well deepens from  $\chi_{\min} = 0.98$  to  $0.44$ .

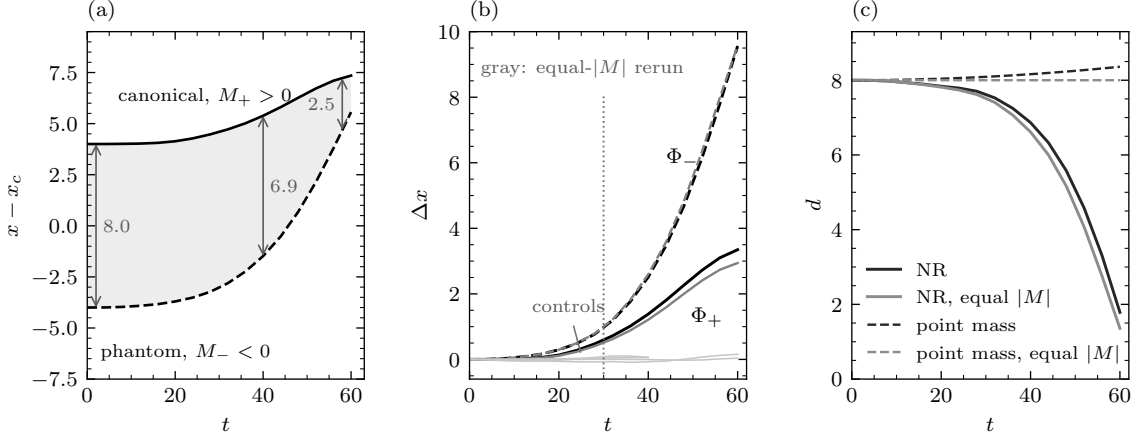


FIG. 3. (a) Barycentre worldlines of run PM: released at rest at  $x = \pm 4$ , both bodies move in  $+x$ , the phantom (dashed) chasing the canonical star (solid); the shaded band is the closing gap. (b) Drifts  $\Delta x$ , solid  $\Phi_+$  and dashed  $\Phi_-$ : black, reference run; gray, PM-eq (Sec. VC); light gray, all four null controls. The dotted line marks the clean-window edge  $t = 30$  (Sec. VI). (c) Separation against the matched point-mass integrations: the NR gap closes to near-contact where the model predicts slow growth.

re-accretes its shed field ( $\chi_{\min}$ :  $0.98 \rightarrow 0.44$ )—whereas the phantom’s own shallow, self-diluting hill ( $\chi \approx 1.001$ ) hands the canonical star no matching boost.

### B. The mirror control closes the artifact objection

A drift produced by grid anisotropy, boundary asymmetry, or diagnostic bias would not know which star is

which; swapping the sectors must reverse the physics but not an artifact. It reverses the physics: flipped in sign, run MP reproduces the reference trajectory to 2.1%/15% (canonical/phantom) at  $t = 30$  and 3.4%/5.2% at  $t = 60$  (Fig. 5).



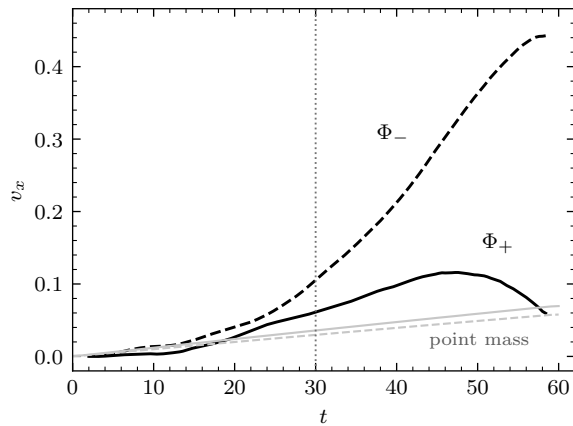


FIG. 4. Barycentre velocities in run PM: centred differences of the continuously sampled barycentre stream over  $\pm 2$  time units (canonical solid, phantom dashed; light gray: the matched point-mass model). The dotted line marks the clean-window edge  $t = 30$ ; beyond it the halo bias (Sec. VI) depresses the canonical curve.

TABLE II. Run PM against a point-mass Bondi integration with the measured ADM masses (each body accelerated by the *other's* active mass). The clean window is  $t \lesssim 30$  (Sec. VI).

| $t$ | drift $\Phi_+$ |       | drift $\Phi_-$ |       | separation |      |
|-----|----------------|-------|----------------|-------|------------|------|
|     | NR             | pm    | NR             | pm    | NR         | pm   |
| 10  | +0.01          | +0.06 | +0.05          | +0.05 | 7.96       | 8.01 |
| 20  | +0.14          | +0.24 | +0.30          | +0.20 | 7.84       | 8.04 |
| 30  | +0.59          | +0.54 | +0.96          | +0.45 | 7.63       | 8.09 |
| 40  | +1.38          | +0.96 | +2.52          | +0.79 | 6.87       | 8.16 |
| 60  | +3.35          | +2.13 | +9.56          | +1.77 | 1.79       | 8.36 |

### C. Gravity scales with the source

The strongest quantitative tests compare runs that differ in exactly one quantity.

*The push scales with the phantom's active mass.* Between PM and PM-eq the phantom's mass changes by the factor 0.831 and nothing else does; the canonical displacement responds in proportion (Table III): drift ratio 0.836 at  $t = 20$ , agreement to 0.6%. Barycentre biases largely cancel in this ratio, since both runs share the same separation and numerics.

*The fall depends only on the other body.* In the same pair of runs the canonical source is unchanged, and the phantom's trajectory with it: drift ratio 1.002 at  $t = 20$  and  $t = 60$  (1.02–1.03 between). Two independent solves and evolutions, differing in the phantom's own mass by 17%, reproduce its trajectory at the percent level—a physics statement and a reproducibility check at once.

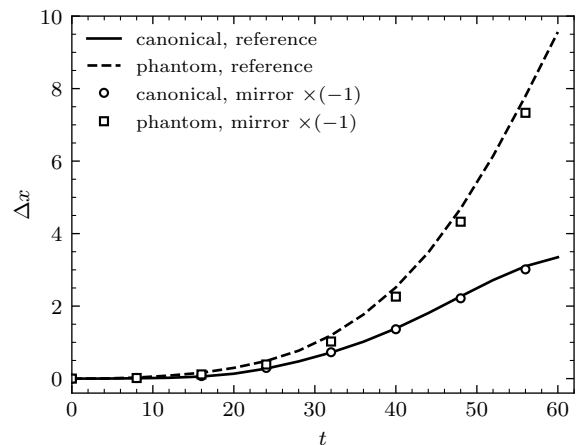


FIG. 5. Mirror control. Lines: reference run PM (canonical solid, phantom dashed); open markers: the mirrored run MP's drift multiplied by  $-1$  (circles canonical, squares phantom). Reflecting the configuration reflects the trajectory to a few percent.

TABLE III. Mass-scaling test: canonical-star drift ratio between PM-eq and the reference run PM, against the same ratio in the point-mass model and the bare mass ratio  $0.06395/0.07696 = 0.831$ .

| $t$ | NR ratio     | point-mass ratio | mass ratio |
|-----|--------------|------------------|------------|
| 20  | <b>0.836</b> | 0.832            | 0.831      |
| 30  | 0.859        | 0.834            | 0.831      |
| 40  | 0.876        | 0.837            | 0.831      |
| 60  | 0.879        | 0.843            | 0.831      |

### D. Separation scaling: inverse-square once the stars separate

Table II shows the reference run departing from the point-mass model: the phantom overshoots increasingly ( $1.5\times$  at  $t = 20$ ,  $3.2\times$  at  $t = 40$ ) and the gap closes where the model grows it. PM-eq isolates the cause: with  $|M|$  matched to  $10^{-5}$  the model predicts identical drifts and a constant gap, yet the NR run gives  $+2.94/+9.59$  and a gap closing to 1.35—not a mass-ratio effect. What remains is geometry: at  $d_0 = 8$  these stars (rms radius  $\approx 5$ ) overlap from  $t = 0$ , the phantom's envelope sitting inside the canonical near field where the pull exceeds  $M/r^2$ .

The wider-separation runs measure this directly (Fig. 6). Fixed-time power-law fits are misleading here—the closing gap makes them report  $n = 2.4$ – $4.6$ , measuring runaway feedback rather than the force law—so each run is compared against its *own* point-mass integration (Table IV). Where the point-mass idealization applies the measured drift is Newtonian to  $\sim 10\%$  at early times; the  $d_0 = 8$  excess is the extended-body overlap enhancement, falling away with separation.

TABLE IV. Ratio of the measured phantom drift to each run’s *own* matched point-mass integration. At  $d_0 = 8$  the stars overlap from  $t = 0$ .

| $d_0$           | $t = 20$ | $t = 30$ | $t = 40$ |
|-----------------|----------|----------|----------|
| 8 (overlapping) | 1.48     | 2.14     | 3.17     |
| 12              | 1.07     | 1.36     | 2.00     |
| 16              | 1.11     | 1.17     | 1.48     |

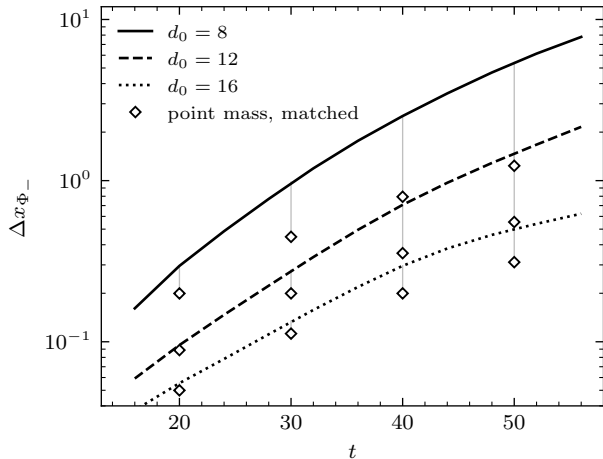


FIG. 6. Separation scaling of the phantom drift (the trustworthy sector; Sec. VI). Curves: NR runs at  $d_0 = 8, 12, 16$ ; diamonds: each run’s *own* matched point-mass integration, tied to its curve at equal times (Table IV).

### E. The phantom sector is the stable one

Counter to intuition, the phantom star was the best-behaved object in the campaign (Table V): gravity is not the binding agent—the identical scalar self-interaction is—and the reversed sign is the *stabilizing* correction. Shed field around the canonical star is pulled back, deepening the well and feeding perturbations ( $\chi$  0.99  $\rightarrow$  0.85 by  $t = 40$ ); around the phantom it anti-gravitates and self-dilutes ( $\chi = 1.0009 \pm 0.0002$  all run). The controls agree: MM keeps  $\chi \approx 1.00$  while its material spreads; PP drives  $\chi$  to 0.44 as it merges.

## VI. CAVEATS, AND THEIR MEASURED SCOPE

*The barycentre includes the halo.* It integrates core plus shed radiation; by  $t = 60$  a third of the tracked canonical field has left the accounting region and the distribution has doubled in extent. The clean quantitative window is  $t \lesssim 30$ , where totals are stable to  $\lesssim 7\%$ —exactly where the mass-scaling and reproducibility tests are strongest. Late-time drifts are upper bounds on core displacement; the qualitative result holds across the whole run.

TABLE V. Single-star health,  $t = 0 \rightarrow 40$ .

| metric                      | canonical                 | phantom                   |
|-----------------------------|---------------------------|---------------------------|
| min $\chi$                  | 0.989 $\rightarrow$ 0.851 | 1.000 $\rightarrow$ 1.001 |
| rms radius (radiation bath) | 5.05 $\rightarrow$ 12.25  | 5.43 $\rightarrow$ 6.25   |
| peak-amplitude breathing    | $\pm 8\%$                 | $\pm 3\%$                 |

*The canonical centroid is biased low.* At  $d_0 = 16$  it drifts *backwards* ( $-0.34$  by  $t = 60$ ): the phantom’s repulsion pushes canonical matter out through the  $+x$  boundary, and losing the leading edge drags the centroid back. Canonical drifts are therefore underestimates everywhere (including the reference  $+3.35$ ), worst where the true displacement is small—wide gaps, early times. The phantom sector, which gains tracked weight rather than losing it, is the trustworthy one, and Sec. VD uses it; the mass-scaling ratio compares equal-separation runs, so the bias largely cancels there. A core-weighted centroid immune to the halo is built for the follow-up.

*The radiation bath cannot leave.* Massive-field radiation reflects from massless-wave (Sommerfeld) boundaries and washes back over the stars; this caps the mixed runs at  $t = 60$ . A sponge layer is the fix.

*Resolution.* Every run used a single resolution ( $N = 128$ ), single rank, effectively unigrid (Appendix A). The only sensitivity probe—the  $30\times$  tighter constraint solve, which changed nothing—addresses the solve, not the evolution; a resolution ladder is future work.

## VII. DISCUSSION AND CONCLUSION

*What is established.* We posed Bondi’s positive-negative mass pair as a full initial-value problem and it behaved exactly as he argued in 1957: released at rest, the mixed pair—and only the mixed pair—self-accelerates, the phantom chasing the canonical star, at 50 to 150 times the noise floor, reversing under mirror reflection. The effect responds to its source like gravity: the push scales with the phantom’s active mass to 0.6%, the fall is set by the canonical star alone and reproduces across independent solves to 0.2%, and the drift converges to the matched point-mass prediction at wide separation. Bondi’s runaway survives the transition from point particles to dynamical, radiating, finite-size matter, and Forward’s momentum accounting survives with it. The dressed phantom star—stable, negative-ADM, constraint-clean—may be the most reusable artifact of the campaign.

*What is not established.* A *constant-gap* runaway—the textbook picture of the pair accelerating forever at fixed separation—requires point-like separation. At  $d_0 = 8$  the stars overlap and close to near-contact by  $t = 60$ ; at  $d_0 = 12\text{--}16$  the gap-closure is milder but the window ends before the asymptotic behaviour is visible. The clean test needs the wider separations *and* a longer window (a

sponge layer), or more compact stars (higher  $\omega$ , where the sector mass asymmetry also shrinks; Fig. 1). Every mixed run was stopped before contact; what a canonical-phantom collision produces—including whether a horizon forms—is untouched. And nothing here bears on whether NEC-violating matter exists; the model is a probe of the gravitational dynamics, not a proposal for a propulsion system [5].

*Next steps.* Core-weighted centroids and a per-sector momentum-balance diagnostic (the direct Bondi book-keeping check) are built and waiting; the separation series should be re-run with them before the scaling numbers are quoted as final. A sponge layer unlocks  $t \gg 60$  and the constant-gap test; the contact end state and a resolution ladder follow.

## REPRODUCIBILITY

All time series, star tables, parameter files, code patches, analysis scripts, and the campaign journal behind every number and figure are packed, with machine paths scrubbed, in the repository’s results tree (`results/bondi-dipole-runaway/`); the figures regenerate from that pack by a single script. Run labels map to pack directories as `PM = pair_pm`, `PM-eq = pair_pm_eqm`, `MP = pair_mp_mirror`, `PP = pair_pp`, `MM = pair_mm`, `P = single_p`, `M = single_m`; the star ODE solver of Sec. III and the  $M(\omega)$  scan behind Fig. 1 are `analysis/star_family_scan.py` and the `boson_star_ode` module it drives. Initial data: `GRtresna` [17]; evolution: `GRteclyn` [16], the AMReX-based [18] GPU port of `GRChombo` [15].

## Appendix A: Constraint behaviour

*Initial data.* `GRtresna`’s nonlinear iteration contracts monotonically on every run; for `PM` the Hamiltonian/momentum error falls 100%  $\rightarrow$  0.083%/0.074% in seven iterations, and every production run exits at  $\leq 0.09\%/\leq 0.08\%$ —the gate quoted in Table I, in the solver’s own grid and normalization.

*Evolution.* The constraints are re-evaluated on the evolution grid every  $\Delta t = 0.5$  and reduced to composite norms over the hierarchy (covered coarse cells dropped, outermost four cells excluded so boundary violation is not misread as interior error). Figure 7 shows the record, in absolute code units: over an overwhelmingly vacuum domain a global *relative* normalization is dominated by

near-zero denominators, so the yardstick is the controls run under identical numerics.

Three features carry the message. The birth transient: re-evaluating with fourth-order stencils after the analytic repaint (Sec. III) gives  $L_2(\mathcal{H}) \simeq 7 \times 10^{-4}$ , relaxing within  $t \lesssim 1$  to the truncation floor  $8 \times 10^{-5}$ . Secular growth:  $L_2(\mathcal{H})$  reaches  $2.7 \times 10^{-3}$  at the window edge  $t = 30$  and  $7.7 \times 10^{-3}$  at  $t = 60$ ,  $L_\infty(\mathcal{H})$  0.05 and 0.9, concentrated

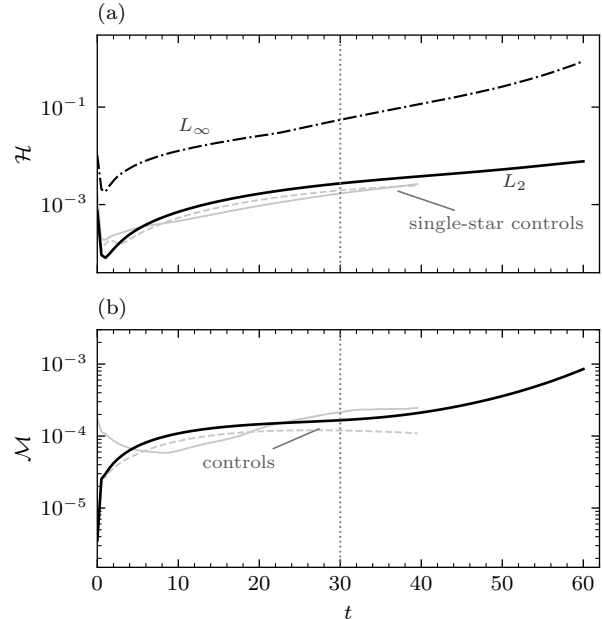


FIG. 7. Constraint monitoring: reference run `PM` (black) against the single-star controls (light gray; solid canonical, dashed phantom). (a) Hamiltonian constraint  $\mathcal{H}$ : composite  $L_2$  norms (solid) and the reference run’s  $L_\infty$  (dash-dotted). (b) Momentum constraint  $\mathcal{M}$ ,  $L_2$ . The equal- $|M|$  rerun lies indistinguishably under the black curves and is omitted; the dotted line marks the clean-window edge  $t = 30$ .

in the deepening canonical well as the pair nears contact and the trapped bath washes back (Sec. VI);  $L_2(\mathcal{M})$  stays below  $2 \times 10^{-4}$  through  $t = 40$ . This growth is the quantitative face of the  $t \lesssim 30$  window. And decisively: through that window the mixed run’s violation is within a factor 1.6 of the non-drifting single-star controls ( $2.7 \times 10^{-3}$  against  $1.7 \times 10^{-3}$  at  $t = 30$ ) while the drifts differ by 50 to 150 times—the drift correlates with the sign structure, not the constraint error.

The single refinement level ( $\chi$ -gradient tagging, regrid every 16 steps) first activates at  $t \approx 47.5$  in the mixed runs and never in the singles: every quantitative result was produced on the uniform  $\Delta x = 0.5$  grid.

- 
- [1] H. Bondi, “Negative mass in general relativity,” *Rev. Mod. Phys.* **29**, 423 (1957).
  - [2] W. B. Bonnor and N. S. Swaminarayan, “An exact solution for uniformly accelerated particles in general relativ-

ity,” *Z. Phys.* **177**, 240 (1964).

- [3] J. Bičák, C. Hoenselaers, and B. G. Schmidt, “The solutions of the Einstein equations for uniformly accelerated particles without nodal singularities,” *Proc. R. Soc. Lon-*



- don A **390**, 397 (1983).
- [4] W. B. Bonnor, “Negative mass in general relativity,” *Gen. Relativ. Gravit.* **21**, 1143 (1989).
  - [5] R. L. Forward, “Negative matter propulsion,” *J. Propulsion Power* **6**, 28 (1990).
  - [6] B. Hoffmann, “Negative mass and the quasars,” in *Perspectives in Geometry and Relativity*, edited by B. Hoffmann (Indiana University Press, Bloomington, 1966), p. 176.
  - [7] R. de A. Martins, “Causal paradoxes implied by the hypothetical coexistence of positive- and negative-mass matter,” *Lett. Nuovo Cimento* **28**, 265 (1980).
  - [8] R. R. Caldwell, “A phantom menace? Cosmological consequences of a dark energy component with super-negative equation of state,” *Phys. Lett. B* **545**, 23 (2002).
  - [9] R. Friedberg, T. D. Lee, and A. Sirlin, “Class of scalar-field soliton solutions in three space dimensions,” *Phys. Rev. D* **13**, 2739 (1976).
  - [10] S. Coleman, “Q-balls,” *Nucl. Phys. B* **262**, 263 (1985).
  - [11] D. J. Kaup, “Klein–Gordon geon,” *Phys. Rev.* **172**, 1331 (1968).
  - [12] R. Ruffini and S. Bonazzola, “Systems of self-gravitating particles in general relativity and the concept of an equation of state,” *Phys. Rev.* **187**, 1767 (1969).
  - [13] S. L. Liebling and C. Palenzuela, “Dynamical boson stars,” *Living Rev. Relativ.* **26**, 1 (2023).
  - [14] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, “Conformal and covariant formulation of the Z4 system with constraint-violating modes,” *Phys. Rev. D* **85**, 064040 (2012).
  - [15] T. Andrade *et al.*, “GRChombo: an adaptable numerical relativity code for fundamental physics,” *J. Open Source Softw.* **6**, 3703 (2021).
  - [16] GRTL Collaboration, *GRTEclyn*, <https://github.com/GRTLCollaboration/GRTEclyn>.
  - [17] J. C. Aurrekoetxea, S. E. Brady, *et al.*, “GRTresna: an open-source code to solve the initial data constraints in numerical relativity,” *arXiv:2501.13046 [gr-qc]* (2025).
  - [18] W. Zhang *et al.*, “AMReX: a framework for block-structured adaptive mesh refinement,” *J. Open Source Softw.* **4**, 1370 (2019).